Tutorial - 8

Q1. If $n(A-B) = 18$, $n(A \cup B) = 70$ and $n(A \cap B) = 25$, find $n(B)$.

Q2. In a Competition, a university awarded medals in different Categories. 36 medals in dance, 12 medals in dramatics and 18 medals in music. If these medals went to a total of 45 persons and only 4 persons got medals in all the three Categories, how many received medals in exactly two of these Categories.

Q3. For each of these Relations on the Set $\{1, 2, 3, 4\}$, decide whether it is Reflexive, whether it is Symmetric whether it is anti-Symmetric and whether it is transitive

(a) $\{(2,2)\ (2,3)\ (2,4)\ (3,2)\ (3,3)\ (3,4)\}$

(b) $\{(1,1)\ (1,2)\ (2,1)\ (2,2)\ (3,3)\ (4,4)\}$

(c) $\{(2,4), (4,2)\}$

(d) $\{(1,2)\ (2,3)\ (3,4)\}$

(e) $\{(1,1)\ (2,2)\ (3,3)\ (4,4)\}$

(f) $\{(1,3)\ (1,4)\ (2,3)\ (2,4)\ (3,1)\ (3,4)\}$

Q4. How many different relations are there from a set with m elements to a set with n elements

Q5. Let R be a relation $R = \{(a,b) \mid a \text{ divides } b\}$ on the set of positive integers. Find

(a) R^{-1} (b) $\overline{R}$

Q6. Let R be the relation on the set of people consisting of pairs (a,b) where a is a parent of b. Let S be a relation on the set of people consisting of pairs (a,b) where a and b are siblings. What are $S \circ R$ and $R \circ S$.

Q7. How many relations are there on a set with n elements that are

(a) Symmetric (b) antisymmetric

(c) asymmetric (d) irreflexive

(e) Reflexive & Symmetric

(f) neither reflexive nor irreflexive

Q8. How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3 \ldots 100\}$ consisting of first 100 positive integers have if R is

(a) $\{(a,b) \mid a > b\}$? (b) $\{(a,b) \mid a \neq b\}$?

(c) $\{(a,b) \mid a = b+1\}$ (d) $\{(a,b) \mid a = 1\}$

(e) $\{(a,b) \mid ab = 1\}$